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Simple methods for low speed calibration of hot-wire anemometers

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ABSTRACT

This paper presents two calibration methods for a CTA probe valid at low speeds covering a range of $0.029\ \mathrm{m/s} \leq U \leq 1.79\ \mathrm{m/s}$. The well-known in situ calibration method using a simple laminar pipe-flow set-up was used for a mean Re number range of $25 \leq Re \leq 1590$. The rotating disc method is the proposed calibration method in the speed range of $0.05\ \mathrm{m/s} \leq U \leq 1.05\ \mathrm{m/s}$. Both calibration methods were compared and found to be in conformity with each other in a maximum error margin of $\pm 7\%$. Therefore derived calibration equation in the form of modified King's Law with corresponding values of the parameters; $A = 2.676$, $B = 0.749$ and $n = 0.75$ can be utilized at speeds as low as 0.029 m/s with sufficient accuracy.

1. Introduction

Hot-wire anemometer is a significant instrument for measurements in laminar, transitional and turbulent flows, due to its accurate interpretation of the signal and simplicity in use. This measuring technique is an indirect method with its output being a voltage signal. Hence, the importance of calibration is a critical manner for effective and accurate measurement of velocity.

The calibration of hot-wire anemometer for high speeds (approximately $U \geq 1.5$ m/s where U is the calibration velocity) can be easily carried out by means of conventional methods using a pitot-static tube and a manometer. However, the calibration of hot-wire anemometer for low velocities (especially, $U < 1.5$ m/s) cannot be carried out using this conventional calibration technique due to the pressure difference (dynamic pressure) in flow being so small that it is hard to read accurately with a manometer.

For low speed calibration of hot-wire anemometers, different techniques have been developed by many scientists. The commonly used ones are laminar pipe-flow method, calibration method based on moving hot-wire anemometer with a known velocity in a stagnant medium and shedding-frequency method. There are also some low speed calibrators and calibration jets manufactured. For example, TSI 1125 calibrator can be used for low speed range between 0.02 m/s and 0.9 m/s. In this calibration jet, the pressure drop from the plenum chamber of the calibration jet is measured and used to obtain velocity from supplied graphs [1].

Kohan and Schwarz [2] used shedding-frequency method and calibrated a hot-wire anemometer at low speeds using Roshko's [3] Strouhal–Reynolds number (SR) relationship for flow Reynolds number ranged between 50 and 150. The velocity was obtained by measuring the vortex-shedding frequency and then using the SR relationship. Christman and Podzimek [4] calibrated a hot-wire using the nozzle of a DISA 55D41/42 calibrator. Aydin and Leutheusser [5] and Tsanis [6] used laminar plane Couette flow and towing tanks as calibration devices to calibrate a hot-wire anemometer at very low velocities capable of operating down to 0.1–0.5 m/s. Bruun et al. [7] carried out a swing-arm apparatus to calibrate hot-wire at low speeds. Lee and Budwig [1] described two methods; laminar pipe-flow method and the shedding-frequency method in the low speed range between 0.15 m/s and 0.95 m/s. Guellouz and Tavoularis [8] used a probe mounted on a pendulum arm for calibration of low speeds. Yue and Malmström [9] used laminar pipe-flow method for hot-wire anemometer in the low speed range of 0.1 m/s and above. Johnstone et al. [10] used the same method to calibrate a hot-wire in the velocity range of 0.125–13.78 m/s. Al-Garni [11] calibrated a hot-wire anemometer at low speed range of 0–0.15 m/s by using a calibration device based on moving hot-wire probes in stagnant air. Durst et al. [12] calibrated hot-wire anemometer by using special calibration test rig including a mass flow control unit and the pressurized container in the range of the mass flow rates per unit area, ρU, of 0.1–25 $\mathrm{kg/m^2 s}$. A comprehensive review on calibration and signal interpretation for hot-wire probes can also be found in [13].

The theoretical approach based on the fluid flow over a heated cylindrical body is used [14,15] in terms of the utilization of the well-known equation;

$$\frac{I^2 R_w}{R_w - R_g} = \frac{E^2}{R_w (R_w - R_g)} = A + BU^n \tag{1}$$

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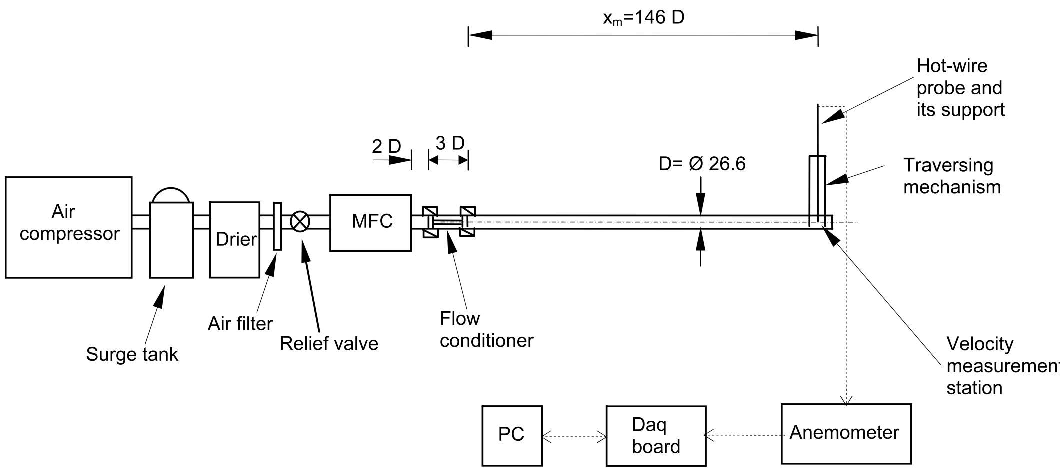

Fig. 1. Schematic diagram of the set-up for laminar pipe-flow method.

where I is current through the hot wire, R_w is resistance of the hot wire and R_g is resistance of the wire at gas temperature, E^2 is square of the output of the hot-wire anemometer bridge, A and B are constants, U is calibration velocity and n is an exponent, usually found to be between 0.45 and 0.5. The most commonly used relationship is the modified King's Law;

$$E^2 = A + BU^n. \tag{2}$$

Besides the modified King's law, extended power law, polynomial model and spline fit models are also present for calibration [16].

In this study, low speed calibration is presented using two methods; in situ calibration using laminar pipe-flow method and a new method based on rotating of a disc with known angular velocity, ω in stagnant air, named as rotating disc calibration method.

2. Set-up and calibration procedure at low air speeds

For both calibration methods, DANTEC 56C01 CTA (constant temperature anemometer) main unit and 56C17 CTA bridge were used. The hot-wire anemometer has uncertainty of less than 5% in resistance measurements. The hot-wire probe used for both calibration methods is a general purpose type Platinum-plated tungsten miniature wire probe, 55P11, which has 5 μm wire diameter and 1.25 mm wire length. The overheat ratio value, OHR was chosen as 1.8 according to the statements given in [15] for both methods. It is the recommended value resulting the maximum wire temperature $T_{w,max}$ as being equal to about 290 °C which is below the oxidization temperature. Since the used probe can only measure one velocity component, the wire element is placed perpendicularly to the mean flow direction. The probe angular position to be perpendicular to the flow direction for both calibration methods was adjusted by rotating the probe around its axis and recording the maximum voltage signal as expressed in [15]. A 16-bit, 1-MHz A/D converter IOtech Daq3001 USB board was used to accumulate and acquisite the raw data for both calibration methods. A sampling frequency of 1 kHz was used for data acquisition. It resulted in a time resolution of 0.001 s. The Daq3001 USB board was set in order to take 1000 readings at a rate of 1 kHz. The temperature of the laboratory was kept constant at 21 ± 0.5 °C during the experiments.

2.1. In situ calibration of hot-wire probe using laminar pipe-flow method

A schematic diagram of the experimental set-up is shown in Fig. 1. In the set-up, air was supplied to the pipeline by means of a LKV 30/8 type of Lupamat screw compressor coupled with a Pnöso type-S PSK-6000F drier and a Pnöso type-PS filter. The compressor can charge the air up to 8 bar with a maximum flow rate of 4.85 m^3/min(0.08 m^3/s). It was used to compress air up to 6 bar through the surge tank, the drier and the filter. The pressure of compressed air was then settled to 4.5 bar by means of a relief valve for safe operation of the mass flow control, MFC unit. The MFC unit can control and measure the air flow rates between 0 and 180 l/min at an accuracy of ±1% as stated by the manufacturer [12,17]. The response time of the displacement controller of the MFC unit is about 2 ms. The test of MFC unit and calibration details are presented in [17]. A flow conditioner composed of a honeycomb, a coarse and a fine screen (Fig. 2a) was used to remove flow non-uniformity at the inlet to the pipeline. The honeycomb was constructed by using plastic straws of 5-mm diameter and 40-mm length ($l/d = 8$ where l is the length of the plastic straw, d its diameter) providing the recommended value of the cell length-to-diameter ratio of honeycombs between 6 and 8 [18,19]. The coarse screen has the screen wire diameter, d_w, of 0.7 mm, the mesh size, M, of 5 mm with an open area ratio, β, of 0.6. The fine screen with $d_w = 0.25$ mm, $M = 1.25$ mm and $\beta = 0.7$ was located 20 mm downstream from the honeycomb in conformity with the recommended distance less than 5 cell sizes [19]. The flow conditioner (Fig. 2b) was connected to the MFC unit by means of a conical custom-designed smooth adapter. Smooth, rigid polyvinyl chloride (PVC) pipeline has inner diameter of 26.6 mm. The minimum required entrance length for fully developed laminar pipe flow at a maximum $Re = 1590$ was found as $x_L = 3.4$ m by means of the following equation [20]:

$$L = 0.08DRe + 0.7D \tag{3}$$

where L is pipe entrance length, D is pipe inner diameter and Re is flow Reynolds number ($Re = U_mD/\nu$ where U_m is mean velocity, ν is kinematic viscosity). The measurement station, x_m, was used as 3.9 m (146 D) downstream of the flow conditioner providing a fully developed flow for all test cases. RCP2-SA6-I-PM-6-200-P1-SBE Robocylinder traverse mechanism with a high positional repeatability of 0.02 mm was used to traverse the hot-wire probe through the cross-section of the pipe.

The working principle of the method is based on the velocity distribution of fully developed laminar pipe flow such that the maximum velocity at the center of the pipe is twice the mean velocity. The hot-wire probe was located at the center of the pipe. The minimum permissible flow rate produced by MFC unit is 0.49 l/min (8.17×10^{-6} m^3/s) with a corresponding mean velocity of 0.0147 m/s. Therefore the calibration velocity at the pipe center is 0.029 m/s being twice the mean velocity of 0.0147 m/s. The

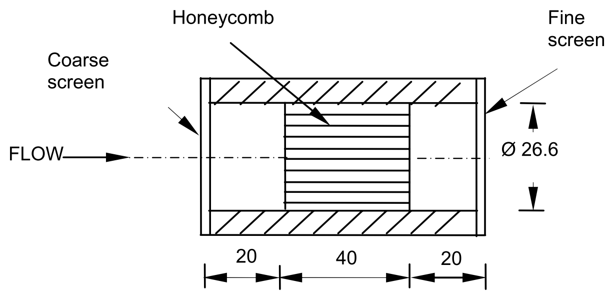

Fig. 2a. Schematic diagram of the flow conditioner.

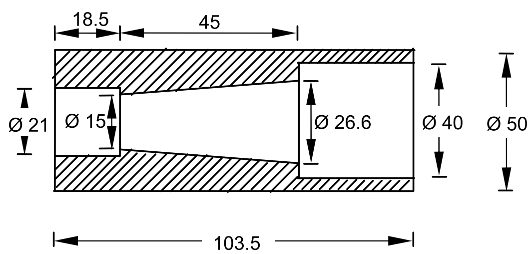

Fig. 2b. Schematic diagram of the conical adapter.

calibration procedure was carried out to the maximum flow rate of 29.89 l/min (4.98×10^{-4} m^3/s) corresponding to the mean velocity of 0.89 m/s, with a calibration velocity of 1.79 m/s at the pipe center. The validity of the method was also checked in terms of the cross-sectional velocity profile measurements for all test cases.

2.2. Design and use of a rotating disc calibration system

A simple system was constructed and used for hot-wire calibration at low speeds similar to the methods based on a moving probe in the stagnant air in the literature [7,8,11]. The calibration set-up comprised of a rotating disc with a known angular velocity, ω, in stagnant air is shown in Fig. 3. The disc was made of steel which had a radius of 9 cm and a thickness of 5 mm. It was driven by a variable-speed AC motor which had a capacity of 1.5 kW. The AC motor was controlled by a SIEMENS micromaster speed control unit (code: 6SE3018-8BC00) which could operate in the angular frequency range of 0 Hz $\leq f \leq$ 650 Hz. A 4-mm diameter hole was drilled on the disc at 8 cm far from the disc center in order to mount the hot-wire probe support. The distance between the disc center and the probe support was selected as 8 cm in order to eliminate the probability of wind effect caused by the disc rotation itself. The probe was placed into the hole and its angular position was adjusted by rotating the probe around its axis and recording the anemometer signal until the maximum voltage signal was obtained as expressed in [15]. The probe cable was attached to the vertical rod on the center of the disc to prevent twist of the cable. In order to disconnect the interaction between the metal support cable end and the vertical rod, rubber rings were used. The calibration set-up was isolated by means of a rectangular box with a volume of 0.94 m^3 having a length of 1.2 m, a width of 1.2 m and a depth of 0.65 m so that the air inside the box was stagnant. The hot-wire probe location was at 0.52 m from the walls of the box eliminating end effects. The probe calibration was free of the effects of rotation in terms of the generated centrifugal force and the probe wake. Since the direction of the centrifugal force was parallel to the axes of 1.25 mm-length of wire which was also between the rigid prongs. Therefore, stretch of hot-wire was negligible and probe wake was prevented by means of balanced disc rotation through insulated driving motor via use of rubber sleeves during the calibration tests. During the calibration, the disc was rotated at angular velocities, ω ranging from 0.63 rad/s to 13.19 rad/s, adjusting the corresponding angular frequency, f, of the motor between 0.1 Hz and 2.1 Hz. The covered angular velocities were calculated using the definition of the angular frequency ($\omega = 2\pi f$). The angular velocities were also determined by using a mechanic tachometer. The angular velocities both calculated from the angular frequency of the motor and by using the tachometer were observed to be the same in a maximum error margin of $\pm 1\%$. At the same time, the hot-wire output voltages for every rotational speeds were recorded to PC by means of Daq3001 USB board. The air speeds sensed by the hot-wire probe were calculated multiplying the angular velocities by the radius of the probe. The calibration range covers the speed range of 0.05–1.05 m/s.

3. Results and discussion

In situ laminar pipe-flow calibration method was performed in the velocity range between 0.029 m/s and 1.79 m/s corresponding to the mean *Re* number range of 25 $\leq Re \leq$ 1590. Rotating disc

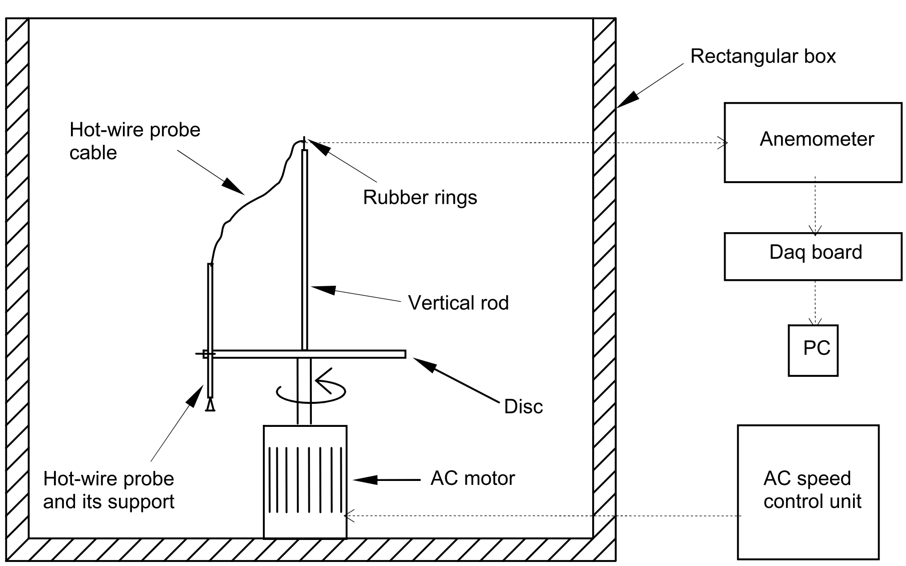

Fig. 3. Schematic diagram of the set-up for rotating disc method.

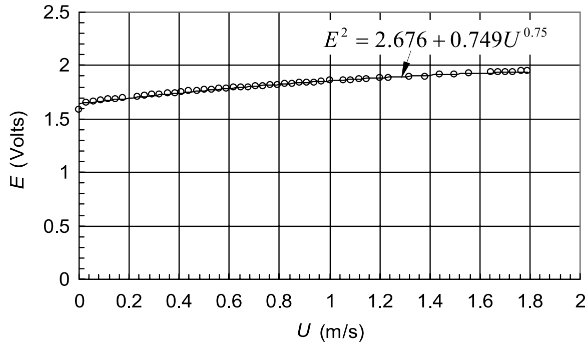

Fig. 4. Low speed calibration curve using laminar pipe-flow method.

calibration method was performed separately from the pipeline set-up in the velocity range between 0.05 m/s and 1.05 m/s. The calibration of the hot-wire probe using rotating disc method was carried out up to the velocity of 1.05 m/s rather than 1.79 m/s. Calibration of the probe for velocities above 1.05 m/s was unsuitable according to the proposed calibration principle due to the observed effects of vibration at high angular velocities of rotating disc.

The laminar pipe-flow method was used for 50 different flow rates with an incremental increase of 8.17×10^{-6} m^3/s. The relationship between the hot-wire voltage outputs and the corresponding velocities using the laminar pipe-flow method is given in Fig. 4. As can be seen from Fig. 4, the calibration curve is in good agreement with the modified King's law in which the A and B constants and n exponent are found as 2.676, 0.749 and 0.75, respectively with a maximum error margin of $\pm 4\%$.

Generally, exponent n is varied between 0.45 and 0.5 for $U \geq 1$ m/s while this exponent increases as the velocity falls to a few centimeters per second [7,8,11]. The exponent n can be seen up to the value of 1 for low speeds [6,11,21]. However, the value of n was found to be 0.75 for the speed range of 0.029–1.79 m/s in the present study.

The fully developed laminar flow character at the measurement station was also verified through the cross-sectional velocity distribution obtained by traversing the hot-wire probe at 35 radial positions at x_m for the covered test cases. The local velocities were calculated from the calibration equation using the hot-wire outputs read at all radial positions. The relationship between the normalized r/R and U/U_{cl} (U_{cl} is the centerline velocity) at a flow rate of 29.89 l/min is given as a sample plot in Fig. 5. The parabolic velocity profile of Blasius is also shown by the dashed lines in Fig. 5. The cross-sectional velocity distribution is in conformity with the parabolic velocity profile of Blasius in a maximum error margin of $\pm 5\%$. The fact of the centerline velocity being twice of the mean velocity was also proved.

The rotating disc method was used for 21 test cases with an incremental increase of 0.0502 m/s. The calibration curve is shown in Fig. 6. As can be seen from Fig. 6, the calibration equation is found to be in good agreement with the modified King's law in which the A and B constants and n exponent are found as 2.625, 0.847 and 0.75, respectively with a maximum error margin of $\pm 5\%$.

In order to evaluate the proposed calibration methods, the velocities were calculated from these methods and the relationships between the calculated and measured velocities were compared for both laminar pipe-flow and rotating disc method as shown in Fig. 7. Two calibration methods have an acceptable confirmation with a maximum deviation of $\pm 7\%$. The maximum deviation of $\pm 7\%$ corresponding to $U \geq 0.8$ m/s may be the result of the aforementioned effects of high angular velocities of rotating disc.

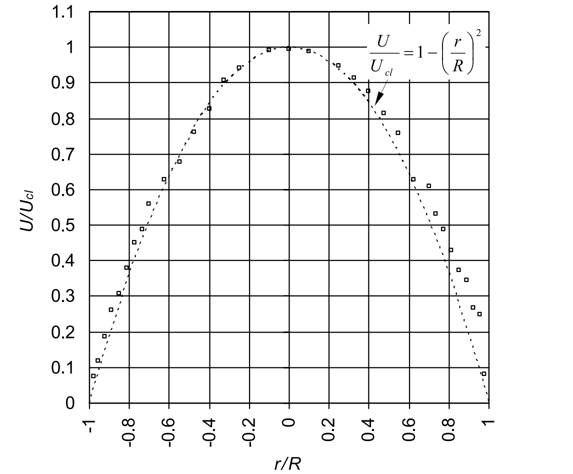

Fig. 5. Cross-sectional velocity distribution at reference flow rate of Q = 29.89 l/min in case of laminar pipe-flow method.

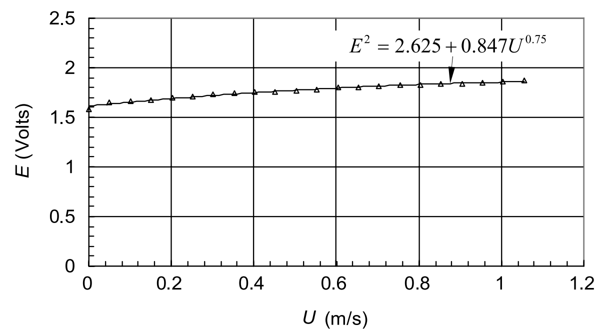

Fig. 6. Low speed calibration curve using rotating disc method.

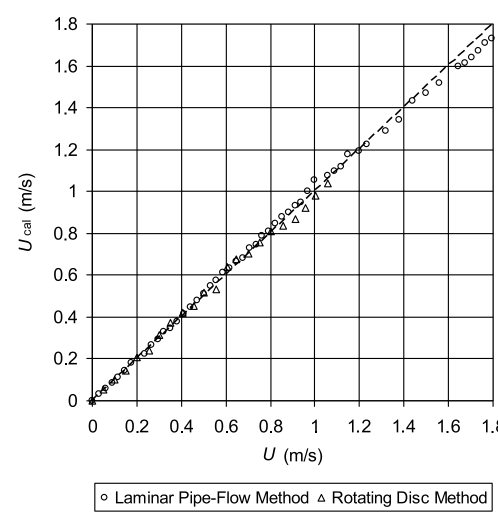

Fig. 7. Comparison of the measured velocities and calculated ones from calibration equations for both laminar pipe-flow and rotating disc method.

4. Conclusion

Two simple, accurate and inexpensive calibration methods of hot-wire anemometers for low speeds are presented in this paper. Rotating disc method contributes to the literature as a new and original low speed calibration method besides in situ laminar pipe-flow method. Both methods are seen to be compatible with each other in an acceptable error margin. The n exponent of modified King's Law, Eq. (2) which can be seen up to the value of 1 for low speeds in the literature was found to be 0.75. Therefore Eq. (2) in which $A = 2.676$, $B = 0.749$ and $n = 0.75$ can be used for calibration of hot-wire probes in the speed range of $0.029\ \text{m/s} \leq U \leq 1.79\ \text{m/s}$. It can be declared that the well-known King's law is convenient also for low speed calibration of hot-wire anemometers as well as its usage for high speed calibration.

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